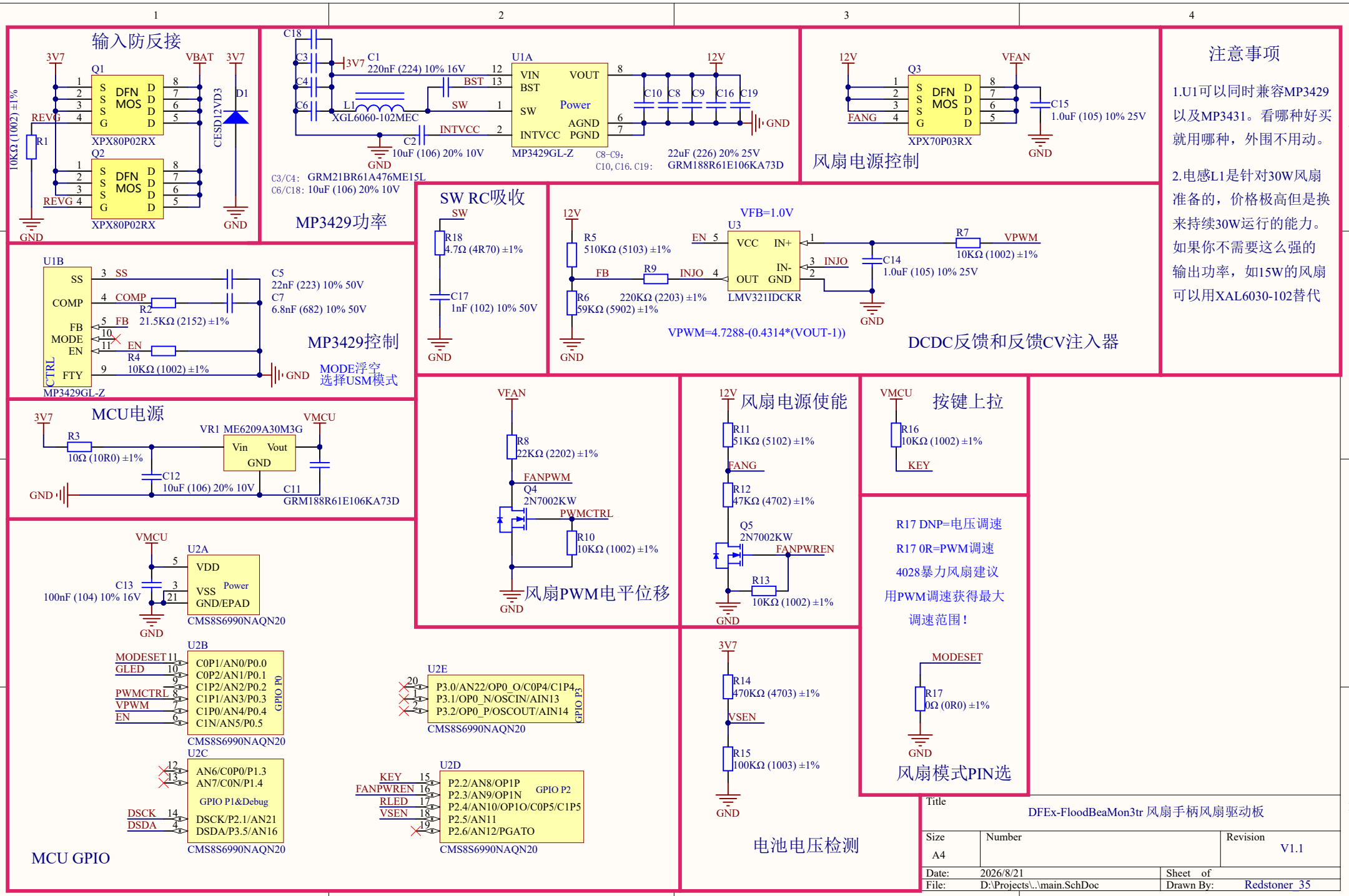
输入防反接
3V7
VBAT
Q1
Q2
DFN MOS
XPX80P02RX
REVG
R1
10KΩ (1002) ±1%
CESD12VD3
D1
GND
C18
C3
C4
C6
C1
220nF (224) 10% 16V
L1
XGL6060-102MEC
C2
10uF (106) 20% 10V
U1A
VIN
BST
SW
INTVCC
VOUT
Power
AGND
PGND
MP3429GL-Z
C10 C8 C9 C16 C19
12V
C8~C9: 22uF (226) 20% 25V
C10, C16, C19: GRM188R61E106KA73D
C3/C4: GRM21BR61A476ME15L
C6/C18: 10uF (106) 20% 10V
MP3429功率
Q3
FANG
C15
1.0uF (105) 10% 25V
VFAN
风扇电源控制
注意事项
1.U1可以同时兼容MP3429以及MP3431。看哪种好买就用哪种，外围不用动。
2.电感L1是针对30W风扇准备的，价格极高但是换来持续30W运行的能力。如果你不需要这么强的输出功率，如15W的风扇可以用XAL6030-102替代
U1B
SS
COMP
FB
MODE
EN
FTY
CTRL
C5
22nF (223) 10% 50V
C7
6.8nF (682) 10% 50V
R2
21.5KΩ (2152) ±1%
R4
10KΩ (1002) ±1%
MODE浮空
选择USM模式
MP3429控制
SW RC吸收
R18
4.7Ω (4R70) ±1%
C17
1nF (102) 10% 50V
VFB=1.0V
U3
VCC
IN+
IN-
OUT
GND
LMV321IDCKR
EN
INJO
R5
510KΩ (5103) ±1%
R6
59KΩ (5902) ±1%
R9
220KΩ (2203) ±1%
C14
1.0uF (105) 10% 25V
R7
10KΩ (1002) ±1%
VPWM
VPWM=4.7288-(0.4314*(VOUT-1))
DCDC反馈和反馈CV注入器
MCU电源
3V7
R3
10Ω (10R0) ±1%
VR1 ME6209A30M3G
Vin
Vout
VMCU
C12
10uF (106) 20% 10V
C11
GRM188R61E106KA73D
VFAN
R8
22KΩ (2202) ±1%
FANPWM
Q4
2N7002KW
PWMCTRL
R10
10KΩ (1002) ±1%
风扇PWM电平位移
12V
风扇电源使能
R11
51KΩ (5102) ±1%
R12
47KΩ (4702) ±1%
Q5
2N7002KW
FANPWREN
R13
10KΩ (1002) ±1%
按键上拉
R16
10KΩ (1002) ±1%
KEY
R17 DNP=电压调速
R17 0R=PWM调速
4028暴力风扇建议
用PWM调速获得最大
调速范围！
MODESET
R17
0Ω (0R0) ±1%
风扇模式PIN选
U2A
VDD
VSS
GND/EPAD
Power
C13
100nF (104) 10% 16V
CMS8S6990NAQN20
U2B
C0P1/AN0/P0.0
C0P2/AN1/P0.1
C1P2/AN2/P0.2
C1P1/AN3/P0.3
C1P0/AN4/P0.4
C1N/AN5/P0.5
GPIO P0
MODESET
GLED
PWMCTRL
VPWM
EN
U2C
AN6/C0P0/P1.3
AN7/C0N/P1.4
GPIO P1&Debug
DSCK/P2.1/AN21
DSDA/P3.5/AN16
DSCK
DSDA
U2E
P3.0/AN22/OP0_O/C0P4/C1P4
P3.1/OP0_N/OSCIN/AIN13
P3.2/OP0_P/OSCOUT/AIN14
GPIO P3
U2D
P2.2/AN8/OP1P
P2.3/AN9/OP1N
P2.4/AN10/OP1O/C0P5/C1P5
P2.5/AN11
P2.6/AN12/PGATO
GPIO P2
KEY
FANPWREN
RLED
VSEN
MCU GPIO
R14
470KΩ (4703) ±1%
R15
100KΩ (1003) ±1%
VSEN
电池电压检测
Title
DFEx-FloodBeaMon3tr 风扇手柄风扇驱动板
Size
A4
Number
Revision
V1.1
Date: 2026/8/21
Sheet of
File: D:\Projects\..\main.SchDoc
Drawn By: Redstoner_35